

Patrick Batsell

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EDUCATION

Rice University

BA in Operations Research and Statistics

Houston, TX

May 2027

- GPA: 3.97
- Coursework: Probability & Statistics, Linear Algebra, Deep Learning, Operations Research & Optimization, Stochastic Modeling, Linear Regression, Computational Thinking, Data Science and Machine Learning

RESEARCH EXPERIENCE

Rice University Electrical and Computer Engineering

Undergraduate Researcher — Supervised by Dr. Richard Baraniuk

Houston, TX

Oct 2025 – Present

- Exploring the Geometry of Training on Synthetic Data.

Rice University Jones School of Business

Undergraduate Researcher — Supervised by Dr. Amit Pazgal

Houston, TX

Feb 2026 – Present

- Working on value-decomposition framework for discrete choice modeling,

Nanyang Technological University

Undergraduate Researcher — Supervised by Dr. Bihan Wen

Singapore

May 2025 – Sep 2025

- Built benchmark for evaluating fine grained classification in Vision Language Models; created synthetic dataset of 8,000+ images with controlled attributes and designed classification tasks to test color, size.

MD Anderson Cancer Center

Undergraduate Researcher — Supervised by Dr. Andrew Schaefer

Houston, TX

Jan 2025 – Feb 2026

- Worked to discover cost parameters for treating *H. pylori*, gastric intestinal metaplasia, and gastric cancer.

PUBLICATIONS

[1] **Patrick Batsell**, Satoshi Tsutsui, and Bihan Wen, “Benchmarking Attribute Discrimination in Infant-Scale Vision-Language Models,” *IEEE International Conference on Image Processing (ICIP)*, 2026; also accepted to the *CVPR 2026 Workshop on Cognitive Foundations for Multimodal Models*.

[2] **Patrick Batsell**, Thomas Walker, and Richard Baraniuk, “A Geometric Perspective on Recursive Synthetic Training,” *ICLR 2026 Workshop on Deep Generative Models*, 2026.

PREPRINTS

[3] **Patrick Batsell**, Thomas Walker, and Richard Baraniuk, “Improving Generative Model Self-training with Geometrically Modified Outputs,” *Under review at ICLR*, 2027.

PROJECTS

Machine Learning Models for Mental Disorder Classification

Houston, TX

- Built and evaluated four models (random forest, gradient-boosted trees, SVM, logistic regression) to identify patients with mental disorders from EEG signals, achieving 85% classification accuracy.

LEADERSHIP & INVOLVEMENT

Rice University Computational and Applied Math Department

Teaching Assistant

Houston, TX

Jan 2026 – Present

- Supported students in class and office hours, guiding them through optimization methods and problem-solving
- Graded assignments and reinforced core concepts through feedback and instruction

Rice University Statistics Department

Teaching Assistant

Houston, TX

Jan 2026 – Present

- Led weekly lab sections and guided students through solving biostatistics problems with code in R

- Graded assignments and designed classes to reinforce core skills.

Rice University Computer Science Department

Teaching Assistant

Houston, TX
Aug 2025 – Dec 2025

- Held weekly office hours and guided students through problem-solving strategies in Python.
- Assisted professors with grading assignments and designing practice problems to reinforce core skills.

Datathon

Director

Houston, TX
Oct 2023 – Feb 2024

- Secured and maintained sponsor relationships, contributing to event funding.
- Oversaw marketing, outreach, and technical support.

SKILLS & INTERESTS

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- **Technical Skills:** Python, R, Gurobi, Machine Learning, Computer Vision, Pytorch, Latex
 - **Languages** English (Native Speaker), Mandarin
 - **Interests:** Water/snow skiing, fitness/weightlifting,

AWARDS

IEEE Signal Processing Society Student Travel Award	September 2026
CVPR Student Travel Award	June 2026
Louise J. Walsh Merit Scholarship	April 2026
President's Honor Roll	2024 - 2026
2nd Place Rice Datathon	February 2025